SPLITTING FILTERED INJECTIVES ALONG STRICT MONOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Two 'Note that' asides (summands stay filtered injective; kernels stay strict monomorphisms) are one-step engine applications or definition expansions, transcribed as-is.

Allowed standard prerequisites (editorial).

- Characterization of strict morphisms. homology-lemma-characterize-strict (cross-chapter, cited in-span; not carded). Supplies the strictness engine used in the kernel induction.

Source reference. Original reference 0127: focused-seeds/geometry/S9-der-split-strict-monomorphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Any strict monomorphism $u : I \to A$ of $\mathit{Fil}^f(\mathcal{A})$ where I is a filtered injective object is a split injection.*

Proof. Let p be the largest integer such that $F^pI \neq 0$. In particular $\mathrm{gr}^p(I) = F^pI$. Let I' be the object of $\mathrm{Fil}^f(\mathcal{A})$ whose underlying object of $\mathcal{A}$ is F^pI and with filtration given by $F^nI' = 0$ for $n > p$ and $F^nI' = I' = F^pI$ for $n \leq p$. Note that $I' \to I$ is a strict monomorphism too. The fact that u is a strict monomorphism implies that $F^pI \to A/F^{p+1}(A)$ is injective, see Homology, Lemma 0127. Choose a splitting $s : A/F^{p+1}A \to F^pI$ in $\mathcal{A}$. The induced morphism $s' : A \to I'$ is a strict morphism of filtered objects splitting the composition $I' \to I \to A$. Hence we can write $A = I' \oplus \mathrm{Ker}(s')$ and $I = I' \oplus \mathrm{Ker}(s'|_I)$. Note that $\mathrm{Ker}(s'|_I) \to \mathrm{Ker}(s')$ is a strict monomorphism and that $\mathrm{Ker}(s'|_I)$ is a filtered injective object. By induction on the length of the filtration on I the map $\mathrm{Ker}(s'|_I) \to \mathrm{Ker}(s')$ is a split injection. Thus we win. □

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